

Q. 1 (Binomial distribution gymnastics). Let $X \sim \text{Binom}(n, p)$, $Y \sim \text{Binom}(m, p)$, and $Z \sim \text{Geom}(p)$ be independent random variables.

- a. Compute $\mathbf{P}\{X + Y = k\}$. Hint: use some probabilistic thinking.
- b. Compute $\mathbf{P}\{X = Z - 1\}$.

Q. 2 (COVID in class). There are 140 students in our class. Every day of the week, each student independently has 0.5% chance of having COVID. As you know, there are lectures on Monday, Wednesday, and Friday.

- a. What is the probability that exactly half the class has COVID during the first lecture?
- b. What is the distribution of the first week in which someone has COVID during at least one of the lectures? (The first week of the semester is week 1, etc.)

Q. 3 (Collecting prizes). Every day, you eat one box of CrunchYum™ breakfast cereal. While this cereal is somewhat disgusting, you are motivated to keep eating it because the manufacturer puts little action figures in the boxes that you like to collect. There are 4 distinct types of action figures. Each box contains an action figure with probability $\frac{1}{2}$; among those boxes that contain an action figure, each type is equally likely. You can assume that the content of every cereal box is independent.

- a. Suppose you have already collected k distinct types of action figures ($0 \leq k < 4$). What is the expected number of boxes you need to eat until you get an action figure of a type that you do not have yet?
- b. From the time that you first start eating CrunchYum™, what is the expected number of boxes you need to eat until you have collected at least one action figure of every type (so you can switch to a healthier breakfast option)?

Q. 4 (Monte Carlo). In this course, we will encounter many models in which we can give exact mathematical formulas for various quantities. In the real world, models are often so complicated that it would be hopeless to give exact mathematical expressions. In such cases, computations must be done numerically on a computer.

One of the most important ideas in this context is the notion of *Monte Carlo methods*. Suppose that Z is some complicated random variable, and we would like to compute

$\mathbf{E}(Z)$. While we cannot do this analytically, it is often possible to program the computer to *simulate* N independent random samples $Z_1, \dots, Z_N$ with the same distribution as Z . The Monte Carlo method approximates the expectation by

$$\mathbf{E}(Z) \approx \hat{Z}_N := \frac{1}{N} \sum_{k=1}^N Z_k.$$

- a. Why is the Monte Carlo approximation a good one for large values of N ?
- b. The *mean square error* of the Monte Carlo approximation is defined as $\text{MSE}(N) := \mathbf{E}((\mathbf{E}(Z) - \hat{Z}_N)^2)$. Suppose that the variance of Z is known to be 1. What is the smallest number of samples N needed to make $\text{MSE}(N) < 0.01$?

Q. 5 (More love triangles). Recall the dating website from the previous homework. There are n people signed up: for every pair of people, the website independently flips a coin that comes up heads with probability p . If it comes up heads, the pair is hooked up. It can happen that three people are all mutually hooked up: this is a love triangle.

In the previous homework, you computed the expected number of love triangles on the entire site. What is the variance of the number of love triangles?